CHE 321 Fu Wu

HW 8

1) $CO + * \rightleftharpoons CO^*$ $\quad k_1 \quad keq_1$

2) $H_2 + 2* \rightleftharpoons 2H^*$ $\quad k_2 \quad keq_2$

3) $CO^* + 2H^* \rightleftharpoons HCOH^* + 2*$ — limiting step $\quad k_3 \quad keq_3$

4) $HCOH^* + * \rightleftharpoons CH^* + OH^*$ $\quad k_4 \quad keq_4$

5) $CH^* + H^* \rightleftharpoons CH_2^* + *$ $\quad k_5 \quad keq_5$

6) $OH^* + H^* \rightleftharpoons H_2O + 2*$ $\quad k_6 \quad keq_6$

overall :

$$CO + H_2 + 2H^* \rightleftharpoons CH_2^* + H_2O + *$$

$$CO + 2H_2 + * \rightleftharpoons CH_2^* + H_2O$$

$$r_1 = -k_1([CO] - keq_1[CO^*])$$
$$r_2 = -k_2([H_2] - keq_2[H^*]^2)$$
$$r_3 = -k_3([CO^*][H^*]^2 - keq_3[HCOH^*])$$
$$r_4 = -k_4([HCOH^*] - keq_4[CH^*][OH^*])$$
$$r_5 = -k_5([CH^*][H^*] - keq_5[CH_2^*])$$
$$r_6 = -k_6([OH^*][H^*] - keq_6[H_2O])$$

since step 3 is limiting step

$$\frac{r_1}{k_1} = \frac{r_2}{k_2} = \frac{r_4}{k_4} = \frac{r_5}{k_5} = \frac{r_6}{k_6} = 0$$

$$\Rightarrow [CO] = [CO^*]keq_1$$
$$[H_2] = [H^*]^2 keq_2$$
$$[HCOH^*] = [CH^*][OH^*]keq_4$$
$$[CH^*][H^*] = [CH_2^*]keq_5$$
$$[OH^*][H^*] = [H_2O]keq_6$$

$$[CO^*] = \frac{[CO]}{keq_1}$$

$$[H^*]^2 = \frac{[H_2]}{keq_2}$$

$$[CH^*] = \frac{[CH_2^*]keq_5}{[H^*]}$$

$$[OH^*] = \frac{[H_2O]keq_6}{[H^*]}$$

$$[HCOH^*] = [CH^*][OH^*]$$

$$[HCOH^*] = keq_4\left(\frac{[CH_2^*]keq_5[H_2O]keq_6}{[H^*]^2}\right)$$

$$r_3 = -k_3\left(\frac{[CO]}{keq_1} \cdot \frac{[H_2]}{keq_2} - keq_3 \cdot \left(keq_4 \frac{[CH_2^*]keq_5[H_2O]keq_6}{\frac{[H_2]}{keq_2}}\right)\right)$$

if $keq_i = 1$

$$r_3 = -k_3\left([CO][H_2] - \frac{[CH_2^*][H_2O]}{[H_2]}\right)$$

2. $2A_{(g)} + D_{(g)} \rightarrow B_{(g)} + C_{(g)}$

overall $-r_A = \dfrac{K P_D P_A^2}{(1+K_A P_A + K_B P_B)^2}$

$r_{ad} = +k_{ad}\left(P_A C_v - \dfrac{1}{K_A} C_{A,s}\right)$

$r_{des} = +k_D\left(C_{B,s} - K_B P_B C_v\right)$

$\dfrac{r_{ad}}{k_{ad}} = \dfrac{r_{des}}{k_D} = 0$

$C_{A,s} = K_A P_A C_v$

$C_{B,s} = K_B P_B C_v$

$A + S \rightleftarrows A\cdot S$

only D, limiting step $\rightarrow 2A\cdot S + D \rightarrow B\cdot S + C + S.$

$B\cdot S \rightleftarrows B + S$

$C_T = C_v + C_{A\cdot S} + C_{B\cdot S}$
$= C_v + K_A P_A C_v + K_B P_B C_v$

reaction step is the limiting step

$C_v = \dfrac{C_T}{(1+K_A P_A + K_B P_B)}$

$-r_A = r_s = k_s (C_{A,s})^2 (P_D)$ ⟷ non reversible

$= k_s\left(K_A P_A \cdot \dfrac{C_T}{1+K_A P_A + K_B P_B}\right)^2 (P_D)$

k_s, K_A, C_T are constant $\Rightarrow K$ (a constant)

$$-r_A = \frac{K P_A^2 P_D}{(1+K_A P_A + K_B P_B)^2}$$